Dynamics of the linear cubic residual
A rank-resolved crossing formula and surviving spectral weight

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Abstract

The energy-orthogonal first coupling-chaos component of the cubic derivative singlet has a nonzero double-scaled dynamical moment. We obtain its exact finite-sector moment by retaining the correlations between the observable and its Hamiltonian. Standard embedded-ensemble covariances reduce the calculation to a terminating Racah polynomial. The normalized second frequency moment tends to $2\mathcal{J}_s^2(1-e^{-4\lambda})/\lambda$, and then to $8\mathcal{J}_s^2$ at small λ. A uniform fourth-moment bound shows that the result is carried by spectral weight that survives at finite frequencies.

1 Physical result and conventions

Use standard CAR, the number-preserving complex Gaussian p-body Hamiltonian $H = H_s$, and coupling covariance $\mathbb{E}J_{IJ}J_{KL} = w\delta_{IL}\delta_{JK}$, including overlapping index sets. The state is infinite temperature and zero chemical potential. Time is conjugate to H_s; $\hbar = 1$ and $\mathcal{J}_s$ has energy units. The sector trace tr_m is normalized by $D_m = \binom{N}{m}$. Write $L = [H, \cdot]$ and

$$T_3 = \tfrac{1}{2}\left(\sum_i c_i^\dagger L^3 c_i + \text{h.c.}\right), \qquad F = \mathcal{W}_1 T_3 - \bar{t}(Q_f)H.$$

Here $\mathcal{W}_1$ means ordinary Gaussian *coupling* chaos, and $\bar{t}$ is the ensemble projection coefficient in each charge sector. The preceding note gives $\langle F^2\rangle_{\mathrm{DS}} = 64\mathcal{J}_s^6 q^2$, where $\lambda = p^2/N > 0$, $q = e^{-4\lambda}$ and $C = \langle H^2\rangle_{\mathrm{DS}} = \mathcal{J}_s^2/\lambda$ [1].

The present result is

$$\boxed{\langle FL^2F\rangle_{\mathrm{DS}} = \frac{128\mathcal{J}_s^8}{\lambda}q^2(1-q), \qquad \mu_{2,F} = \frac{\langle FL^2F\rangle}{\langle F^2\rangle} = \frac{2\mathcal{J}_s^2}{\lambda}(1-q).} \tag{1}$$

In particular $\mu_{2,F} \to 8\mathcal{J}_s^2$ after the DS limit. The corresponding previously obtained values for X and R are $16\mathcal{J}_s^2$ and $24\mathcal{J}_s^2$. This motivates testing a q-probe kernel for F at all orders. This note establishes the curvature and surviving dynamics; the complete kernel is the next calculation.

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2 Exact rank covariance

Initially take $p \le \min(m, N-m)$. In the standard multiplicity-free $U(N)$ decomposition, $H = \sum_{a=0}^{p} H_a$ and the Gaussian H_a are independent. Define

$$C_r = \binom{m}{r}\binom{N-m+r}{r}, \qquad d_a = \binom{N}{a}^2 - \binom{N}{a-1}^2, \tag{2}$$

$$\omega_a^{(r)} = \frac{d_a \binom{m-a}{m-r}\binom{N-r-a}{m-r}}{D_m C_r}, \qquad \sum_{a=0}^{r} \omega_a^{(r)} = 1. \tag{3}$$

The physical variance is $C_m = wC_p$, and $v_a = \mathbb{E}\,\mathrm{tr}_m\, H_a^2 = C_m \omega_a^{(p)}$. For an irreducible tensor Y_b, introduce the normalized map

$$\mathbb{E} H_a Y_b H_a = v_a g_{ab} Y_b. \tag{4}$$

The known full r-body covariance gives [2, 3]

$$\sum_{a=0}^{r} \omega_a^{(r)} g_{ab} = \frac{\binom{m-b}{r}\binom{N-m+r-b}{r}}{C_r}. \tag{5}$$

Since the diagonal weight is positive, this triangular system uniquely specifies the pure-rank map. Its explicit solution is

$$\boxed{g_{ab} = {}_4F_3\left(\begin{matrix}-a, a-N-1, -b, b-N-1 \\ -m, m-N, -N-1\end{matrix}; 1\right).} \tag{6}$$

The sum ends at $\min(a, b)$, before any denominator vanishes. It is a standard Racah polynomial, with parameters $(\alpha, \beta, \gamma, \delta) = (-m-1, m-N-1, -N-2, 0)$ in DLMF 18.26.3 [5]. No general positive-parameter orthogonality formula is required here.

For completeness, let $z_k(a) = (-a)_k (a-N-1)_k$. Direct substitution of (3) and the terminating Rogers–Dougall sum gives

$$\sum_a \omega_a^{(r)} z_k(a) = \frac{(-r)_k (m-N)_k (-N-1)_k}{(m-N-r)_k}. \tag{7}$$

One explicit way to check the summation is to write $A = -N-1$ and

$$\frac{\omega_a^{(r)}}{\omega_0^{(r)}} = \frac{A+2a}{A} \frac{(A)_a^2 (-r)_a (m-N)_a}{(a!)^2 (-m)_a (r-N)_a}.$$

After multiplying by $z_k(a)$ and shifting $a = k+j$, the very-well-poised parameters are $A' = A+2k$, $b' = A+k$, $c' = m-N+k$, $d' = -(r-k)$. Apply DLMF 16.4.9 in its terminating, rational form [4]. Inserting (7) into (6) leaves ${}_3F_2(-b, b-N-1, -r; -m, m-N-r; 1)$. Pfaff–Saalschutz, DLMF 16.4.3, gives exactly (5). This proves the stated covariance solution.

3 Same-disorder Wick contractions

The previous finite-sector result is $\mathcal{W}_1 T_3 = \sum_a t_a H_a$, so set

$$\bar{t} = \sum_a \omega_a t_a, \qquad f_a = t_a - \bar{t}, \qquad F = \sum_a f_a H_a, \qquad V_F = \sum_a v_a f_a^2. \tag{8}$$

Here and below $\omega_a = \omega_a^{(p)}$. The coefficients t_a, including their factor w, are fixed by the formula and code in [1]; no fit or independence assumption is made. In particular $\sum_a v_a f_a = 0$ and $\mathbb{E}FH = \mathbb{E}HF = 0$ as matrices.

Let $B_{20} = \mathbb{E}\,\mathrm{tr}_m\, F^2H^2$ and $B_{11} = \mathbb{E}\,\mathrm{tr}_m\, FHFH$. All three Gaussian pairings give

$$B_{20} = C_m V_F + \sum_{a,b} v_a v_b f_a f_b g_{ab}, \tag{9}$$

$$B_{11} = \sum_{a,b} v_a v_b f_a^2 g_{ab}. \tag{10}$$

The other noncrossing pairings vanish because $\sum v_a f_a = 0$. The second term of (9) is the same-disorder contraction that an independent-probe substitution would omit. By cyclicity $\langle FL^2F\rangle = 2(B_{20} - B_{11})$. The symmetry $g_{ab} = g_{ba}$, manifest in (6), therefore yields

$$\boxed{\langle FL^2F\rangle_m = \sum_{a,b} v_a v_b (f_a - f_b)^2 (1 - g_{ab}).} \tag{11}$$

This also follows from $[H, F] = \sum_{a<b}(f_b - f_a)[H_a, H_b]$. The map in (4), divided by v_a, is completely positive, unital and trace preserving. Its Hilbert–Schmidt contraction property gives $|g_{ab}| \le 1$. Thus every summand in (11) is nonnegative. Adding a common constant to all t_a leaves the dynamic moment unchanged. The scalar channel has $g_{0b} = 1$ and contributes no commutator.

An exact small-system example, with $w = 1$, is

$$(N, p, m) = (5, 2, 3): \quad C_m = 18, \quad V_F = 180, \quad B_{20} = 3420, \quad B_{11} = 1800, \quad \langle FL^2F\rangle = 3240. \tag{12}$$

Here the extra term in (9) is 180, so it cannot be discarded at finite size. This is an ensemble moment, distinct from the earlier deterministic noncommutation witness.

For evaluation one need not perform a cubic-cost rank sum. With $c_a = t_a/C_m$, $\delta_a = c_a - \sum \omega c$, define

$$A_{j,k} = \sum_a \omega_a \delta_a^j z_k(a), \qquad h_k = \frac{1}{(-m)_k (m - N)_k (-N - 1)_k k!}.$$

Then the two extra contractions, divided by C_m^4, are $\sum_k h_k A_{1,k}^2$ and $\sum_k h_k A_{2,k} A_{0,k}$, respectively. These finite sums retain all ranks and signs.

4 Taking the double-scaled limit

At half filling let $a = p - s$, $b = p - u$, with fixed nonnegative s, u, and hold $p^2/N = \lambda > 0$. The kth summand T_k in (6) satisfies

$$\frac{T_{k+1}}{T_k} = -\frac{(a-k)(N+1-a-k)(b-k)(N+1-b-k)}{(m-k)(N-m-k)(N+1-k)(k+1)}.$$

For each fixed k this tends to $-4\lambda/(k+1)$. For all allowed k, sufficiently large N gives $|T_{k+1}/T_k| \le 16\lambda/(k+1)$, by bounding $a, b \le p$ and all three denominator factors away from zero. The factorial majorant permits termwise summation:

$$g_{p-s,p-u} \longrightarrow \sum_{k\ge 0} \frac{(-4\lambda)^k}{k!} = q. \tag{13}$$

This step controls the *pure-rank* covariance, not just the average over ranks in (5).

The preceding note proves, with polynomial-moment convergence,

$$s \implies \text{Poisson}(\lambda), \qquad c_{p-s} - c_p \longrightarrow 8\lambda q s, \qquad \text{Var}_\omega\, c \longrightarrow W = 64\lambda^3 q^2. \tag{14}$$

Its bounds $\omega_{p-s} \le K_\lambda (32\lambda)^s/s!$ and $|c_{p-s} - c_p| \le K_\lambda(1+s)$ are enough here. In particular $|g_{ab}| \le 1$ controls the tails of both rank sums, including their coefficient differences. Combining (11)–(14) gives

$$\frac{\langle FL^2F\rangle_m}{C_m^4} \longrightarrow (1-q)\mathbb{E}\Big[(8\lambda q(s-u))^2\Big] = 128\lambda^3 q^2(1-q), \tag{15}$$

where s, u are independent Poisson variables in this sum. Since $V_F = C_m^3 \,\text{Var}\, c$, (1) follows.

The same argument supplies two separate energy moments:

$$\frac{B_{20}}{C_m V_F} \to 1, \qquad \frac{B_{11}}{C_m V_F} \to q. \tag{16}$$

The same-disorder term tends to zero because its limiting value is $q(\mathbb{E}\delta)^2 = 0$, *after* the covariance and rank-tail limits have been justified.

For the full infinite-temperature trace, the charge distribution is $2^{-N}\binom{N}{m}$. The preceding concentration estimates are uniform for $m/N \to 1/2$. For example restrict first to $|m - N/2| \le N^{2/3}$, use the same rank and hypergeometric bounds there, and then remove the exponentially small complement. The rough global bounds $|t_a| \le KNC_{\max}$ and $C_{\max}/C_{N/2} \to 1$ control that complement. Fixed-sector formulas outside $p \le \min(m, N-m)$ are not needed for this tail bound. The full-trace limits in (1) and (16) follow. The order is DS first, small λ second.

5 Why a nonzero moment represents surviving dynamics

The normalized positive frequency measure in sector m is

$$P_N(d\omega) = \frac{1}{D_m V_F} \mathbb{E} \sum_{i,j} |F_{ij}|^2 \delta_{E_i - E_j}(d\omega). \tag{17}$$

It is symmetric, has total mass one, and its second moment is $\mu_{2,N}$. To rule out a vanishing amount of weight escaping to infinite frequency, we bound its fourth moment uniformly.

For a Hermitian Gaussian matrix series Y with $\mathbb{E}Y^2 = s^2\mathbf{1}$, integration by parts gives

$$\mathbb{E}\,\mathrm{tr}_m\, Y^{2k} \le (2k-1)!!\, s^{2k}. \tag{18}$$

A short proof is useful. Write $Y = \sum x_\alpha A_\alpha$, $\sum A_\alpha^2 = s^2\mathbf{1}$, and differentiate $\mathrm{tr}\, A_\alpha Y^{2k-1}$. Each of the $2k-1$ resulting terms, summed over α, is bounded in absolute value by $s^2\,\mathrm{tr}\,|Y|^{2k-2}$. In an eigenbasis of Y this follows from weighted Young's inequality applied to $|y_i|^\ell |y_j|^{2k-2-\ell}$ and $\sum_{\alpha,j} |(A_\alpha)_{ij}|^2 = s^2$. Induction proves (18), without independence between H and F.

Apply this separately to H and F; the latter is also a Gaussian series and $\mathbb{E}F^2 = V_F\mathbf{1}$ by $U(N)$ invariance. Cauchy–Schwarz in ensemble times normalized trace gives

$$\mu_{4,N} \le \frac{16}{V_F} \mathbb{E}\,\mathrm{tr}_m\, F^2 H^4 \tag{19}$$

$$\le \frac{16}{V_F} \sqrt{\mathbb{E}\,\mathrm{tr}_m\, F^4\; \mathbb{E}\,\mathrm{tr}_m\, H^8} \le 16\sqrt{315}\, C_m^2. \tag{20}$$

The first inequality uses $(x-y)^4 \le 8(x^4+y^4)$. Thus P_N is tight and its second moments are uniformly integrable. Every subsequential DS limit P has

$$\int \omega^2 P(d\omega) = 2C(1-q) > 0.$$

It cannot be concentrated at zero frequency. More quantitatively, Paley–Zygmund applied to ω^2 yields

$$P\Big(|\omega| \ge \sqrt{C(1-q)}\Big) \ge \frac{(1-q)^2}{16\sqrt{315}} > 0. \tag{21}$$

This conservative bound is for fixed $\lambda > 0$. It proves surviving dynamics, not absence of an additional static atom. The full-trace version follows by the same central charge window; the Gaussian bounds and polynomial bounds on t_a suppress the exterior contribution as well.

The normalized correlator consequently has the uniform local expansion $C_{F,N}(t) = 1 - \mu_{2,N} t^2/2 + O(\mu_{4,N} t^4)$, with remainder at most $\mu_{4,N} t^4/24$. The quadratic coefficient matches a q probe. Establishing a unique complete limiting measure requires higher two-energy moments; the mixed correlators with R, S, Z are also still to be computed. Neither curvature nor the rank label specifies a bulk mass.

6 Verification, provenance and continuation

The exact covariance and special-function identities used here are established mathematical tools [2, 3, 4, 5]. Our calculation applies them to the previously constructed same-disorder rank filter F. We do not claim a new Racah polynomial identity or an independent discovery of embedded-ensemble covariances.

The saved checks are:

- 1,515 exact rational comparisons of the triangular covariance inverse with (6), 928 factorial-moment identities, and symmetry and contraction bounds for $3 \le N \le 14$.
- Five finite CAR sectors in $(N, p) = (3, 1), (4, 2), (5, 2)$. Direct cubic polarization and the complete quadratic Gaussian Wick contraction reproduce (11), including the nonzero value (12). Matrix entries are bounded Gaussian integers and final sums are exact rational numbers.
- Three separate CAR checks of all three pairings of each of B_{20} and B_{11}, and six comparisons of the accelerated finite sum with exact rational double-rank sums.
- Twelve full-rank DS calculations, $\lambda = .1, .25, .5$ and $p = 16, 32, 64, 128$, with 80 and 120 decimal working digits. The 65-digit stored normalized moments agree; this precision comparison is not interval certification.

The numerical convention sets the finite-sector variance exactly to $C_m = \mathcal{J}_s^2/\lambda$ and $\mathcal{J}_s = 1$. It therefore isolates the dimensionless finite-rank correction from a normalization change. At $p = 128$ the results are

λ	finite μ_2	DS μ_2	relative deviation
0.10	6.55161515	6.59359908	−0.63674%
0.25	5.03380082	5.05696447	−0.45805%
0.50	3.45001236	3.45865887	−0.25000%

All ranks and every terminating-series term are included. The alternating contraction sums and their cancellation factors are recorded in the JSON output. Dependencies and input/output hashes are stored with the local evidence. No Git commit or push was made.

The next task is to calculate $\langle F H^u F H^v \rangle$ beyond total degree two, retaining the H–F pairings. A full q-kernel theorem would then permit a direct comparison of this channel with the known q^2 and q^3 singlets, before combining the raw cubic operator or constructing its mixed interactions.

References

[1] A. Day and Chengheng, *The missing linear component of the cubic singlet*, local note, 6 November 2026, `manuscript/susskind-cubic-linear/main.tex`.

[2] V. K. B. Kota, *Bivariate moments of the two-point correlation function for embedded Gaussian unitary ensemble with k-body interactions*, arXiv:2208.11312v2, Sec. IV, Eqs. (28)–(31), (37)–(39).

[3] J. Tall and S. Tomsovic, *Double-scaled bosonic and fermionic embedded ensembles, complex SYK, and the dual Hilbert space*, arXiv:2604.14522v2, Sec. 2.1, especially Eq. (2.12).

[4] NIST DLMF, 16.4.3 and 16.4.9, terminating Pfaff–Saalschutz and Rogers–Dougall sums.

[5] NIST DLMF, 18.26.3, Racah polynomial representation.